
AP PHYSICS C

Mechanics and Electromagnetism

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0.1 Foreword for anybody else using this

As of writing this, I've already taken AP Physics 1 and 2 and Calc BC, so I'll skim over those stuff. I steal the images from google. If this violates copyright somehow, please email me and I'll remove it.

1 Kinematics

1.1 Scalars and Vectors

Scalars are numbers. Vectors are vectors. Represent as $\vec{a} = \hat{i} + \hat{j} + \hat{k}$.

Dot product of two vectors is the length of one of the vectors projected onto the other multiplied by the length of said other vector. Theta = min angle between two vectors.

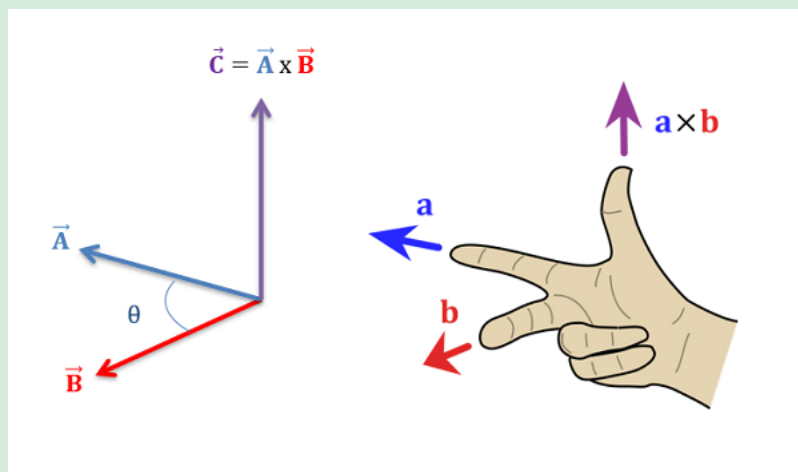
$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$$

1.1.1 Cross Product

Operation that can be applied to two **three-dimensional** vectors to generate a third vector perpendicular to both vectors. Use the **right hand rule** to determine which direction the third vector will go.

Principle 1.1 (Right Hand Rule)

To find the geometrical direction of $\vec{a} \times \vec{b}$, hold your right hand out and point your *index* finger in the direction of $\vec{a}$, and your *middle* in the direction of $\vec{b}$. Your thumb will point in the direction of $\vec{a} \times \vec{b}$.



Principle 1.2 (Calculating the Cross Product)

To calculate the cross product between $\vec{A}$ and $\vec{B}$:

EX. Compute $\langle 1, 0, -1 \rangle \times \langle 2, 3, -1 \rangle$.

$\hat{i}$	$\hat{j}$	$\hat{k}$	$\hat{i}$	$\hat{j}$
1	0	-1	1	0
2	3	-1	2	3

$= 0(-1)\hat{i} + -1(2)\hat{j} + 1(3)\hat{k}$
 $- 3(-1)\hat{i} - (-1)(1)\hat{j} - 2(0)\hat{k}$
 $= 3\hat{i} - \hat{j} + 3\hat{k}$
 $= \langle 3, -1, 3 \rangle$

Note the following:

1. $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$ where θ is the minimum angle between the two vectors. Also equal to the area of the parallelogram formed by $\vec{a}$ and $\vec{b}$.
2. $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. This property is known as the **anti-commutativity of the cross product**.
3. $\vec{a} \times \vec{a} = \vec{0}$ ($\theta = 0$)

1.2 Displacement, Velocity, and Acceleration

$$\vec{v} = \frac{d\vec{s}}{dt}, a = \frac{d\vec{v}}{dt} = \frac{d^2\vec{s}}{dt^2} \implies \Delta\vec{s} = \int \vec{v}(t)dt, \Delta\vec{v} = \int \vec{a}(t)dt$$

Derivatives and integrals of vector-valued functions can be done component wise.

$$\vec{s} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k} \implies \frac{d\vec{s}}{dt} = x'(t)\hat{i} + y'(t)\hat{j} + z'(t)\hat{k}$$

$$\vec{s} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k} \implies \int_{t_i}^{t_f} \vec{s}dt = \int_{t_i}^{t_f} x(t)dt\hat{i} + \int_{t_i}^{t_f} y(t)dt\hat{j} + \int_{t_i}^{t_f} z(t)dt\hat{k}$$

For constant one-dimensional acceleration, kinematic equations apply:

$$\Delta v = \int a dt \implies v_f = v_i + at$$

$$\Delta s = \int v(t)dt = \int (v_i + at)dt \implies \Delta s = v_i t + \frac{1}{2}at^2$$

$$v_f = v_i + at \implies v_f^2 = v_i^2 + 2v_i at + a^2 t^2 \implies v_f^2 = v_i^2 + a(2v_i t + at^2) \implies v_f^2 = v_i^2 + 2a\Delta x$$

Falling object experience constant acceleration: $a = g \approx 9.81\text{m s}^{-2} \approx 10\text{m s}^{-2}$. Projectiles experience constant vertical acceleration of gravity, but no horizontal acceleration.

1.3 Representing Motion

Distance, velocity, and acceleration vs. time graphs exist. Yipee!

1.4 Reference Frames and Relative Motion

Let $\vec{v}_{AB}$ = velocity of A relative to B.

$$\vec{v}_{AB} = -\vec{v}_{BA}$$

$$\vec{v}_{AC} = \vec{v}_{AB} + \vec{v}_{BC}$$

These are also known as **Galilean Transformations**. Note these equations will break down near the speed of light due to relativity shit.

Example 1.3

The president's airplane flies at 250m s^{-1} to the east with respect to the air. The air is moving at 35m s^{-1} to the north with respect to the ground. Find the velocity of the president's airplane with respect to the ground.

Answer 1

Let $\hat{i}$ = north and $\hat{j}$ = east.

$$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG} = (250\hat{j} + 35\hat{i})$$

$$|\vec{v}_{PG}| = \sqrt{250^2 + 35^2} \approx 252.4 \text{ m s}^{-1}$$

$$\theta_{PG} = \arctan \frac{35}{250} \approx 7.97^\circ \text{ relative to East}$$

1.5 Motion in Two or Three Dimensions

everything that we've talked about already considers three dimensions :3

2 Force and Translational Dynamics

2.1 Systems and Center of Mass

Recall that the location of a system's center of mass along a given axis can be calculated as follows:

$$\vec{x}_{cm} = \frac{\sum m_i \vec{x}_i}{\sum m_i}$$

This equation can be extended for non-uniform solids by considering differential masses dm :

$$\vec{r}_{cm} = \frac{\int \vec{r} dm}{\int dm}$$

One way to calculate dm is through **linear mass density**:

$$\lambda = \frac{dm}{dl}$$

Area and volume mass density exists as well, represented by σ and ρ respectively.

Example 2.1

Suppose you have a bar of length L which has mass density function $\lambda(r) = cr^2$. Find the center of mass in terms of L , c , and fundamental constants.

Answer 2

$$\frac{dm}{dr} = cr^2 \implies dm = cr^2 dr$$

$$\begin{aligned} r_{cm} &= \frac{\int r dm}{\int dm} \\ &= \frac{\int_0^L cr^3 dr}{\int_0^L cr^2 dr} \\ &= \frac{\frac{r^4}{4} \Big|_0^L}{\frac{r^3}{3} \Big|_0^L} \\ &= \frac{3}{4} \frac{L^4}{L^3} \\ &= \boxed{\frac{3}{4} L} \end{aligned}$$

Also recall that for force diagrams, systems can be modeled as a single point located at the system's center of mass.

2.2 Forces and Free-Body Diagrams

they exist

2.3 Newton's Third Law

$$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$$

Recall that tension is the macroscopic net of forces segments of a string, cable, chain or a similar system exerts on each other. The tension has negligible mass, does not stretch, and has equal tension across all points in the string.

2.4 Newton's First Law

It's called **translational equilibrium** if the net force on the system is zero.

2.5 Newton's Second Law

$$\vec{F}_{net} = m_{sys}\vec{a}_{sys}$$

2.6 Gravitational Force

Definition 2.2 (Newton's law of universal gravitation)

$$|\vec{F}_g| = G \frac{m_1 m_2}{r^2}$$

The gravitational force is always exerted along the line connecting the center of masses of two systems.

This force suggests that a mass sets up a **gravitational field** around itself: $|\vec{g}| = G \frac{M}{r^2}$. The word **weight** is only used when one of the objects is much more massive than the other (usually the Earth). **Apparent weight** is the normal force acting on an object, and is what is measured by a scale in an elevator. Near the surface of the earth, $g \approx 10 \text{ N kg}^{-1}$.

Newton's Shell Theorem states *the net gravitational force exerted on an object inside a thin spherical shell is zero AND the net gravitational force exerted on an object by a spherical shell is equivalent to treating the shell as a point mass.*

In practice, the first half of Newton's Shell Theorem suggests that **within a uniform sphere, only the mass of a sphere within a radius equal to the object's distance from the center** affects it. All outer layers can be considered as differential spherical shells, and by Newton's Shell Theorem exert no gravitational force. Thus,

$$m_{partial} = \rho \frac{4}{3} \pi (r_{partial})^3 \text{ (this equation is NOT on the sheet)}$$

Using the second half of Newton's Shell theorem and the law of gravitation:

$$\begin{aligned} F_g &= \frac{Gm}{r_{partial}^2} * \rho \frac{4}{3} \pi (r_{partial})^3 \\ &= \frac{4}{3} G m \rho \pi r_{partial} \\ \implies F_g &\propto r_{partial} \text{ (linear proportionality within the sphere!)} \end{aligned}$$

Example 2.3

Find the weight of a 1.0 kg mass located half way to Earth's center. Ignore density variation.

Answer 3

$$F_g \propto r_{\text{partial}} \implies F_g = \frac{mg}{2} = \frac{1.0 * 10}{2} = \boxed{5\text{N}}$$

2.7 Kinetic and Static Friction

Frictional force is exerted to oppose a change in motion on a rough surface, and is proportional to the magnitude of the normal force. The proportionality constant depends on whether it is moving (**kinetic**) or stationary (**static**). Kinetic friction is usually less than static friction.

2.8 Spring Forces

$$\vec{F}_{\text{spr}} = -k\Delta x$$

$$k_{\text{eq,parallel}} = \sum k_i$$

$$\frac{1}{k_{\text{eq,series}}} = \sum \frac{1}{k_i}$$

2.9 Resistive Forces

A resistive force is defined as a **velocity-dependent** force that opposes an object's velocity.

$$\vec{F}_{\text{res}} = -k\vec{v}$$

Setting up a second law equation with a resistive force will lead to a differential equation for velocity (recall $a = \frac{dv}{dt}$), which can be solved to lead to an exponential equation for velocity. Terminal velocity is defined as the maximum velocity reached under a resistive force.

Example 2.4

A skydiver of mass m drops from rest, and experiences air resistance $F_{\text{air}} = -bv$. Assuming they do not hit the ground, find an equation for their velocity as a function of time. State what this suggests about their terminal velocity.

Answer 4

$$\begin{aligned}F_{net} &= mg - F_{air} \\m \frac{dv}{dt} &= mg - bv \\ \frac{m}{mg - bv} dv &= 1 dt \\ \int \frac{m}{mg - bv} dv &= \int dt \\ -\frac{m}{b} \ln(mg - bv) + C &= t \\ \text{init cond } v = 0, t = 0 \\ -\frac{m}{b} \ln(mg) + C &= 0 \\ \implies C &= \frac{m}{b} \ln(mg) \\ -\frac{m}{b} \ln\left(\frac{mg - bv}{mg}\right) &= t \\ \frac{mg - bv}{mg} &= e^{-\frac{bt}{m}} \\ \implies v &= \frac{mg}{b} (1 - e^{-\frac{bt}{m}})\end{aligned}$$

$$\text{As } t \rightarrow \infty, \boxed{v \rightarrow \frac{mg}{b}}$$

2.10 Circular Motion

In order to maintain a circular path of radius r , a particle needs to experience $a_c = \frac{v^2}{r}$ directed towards the center of the motion. This is known as **centripetal acceleration**. The acceleration along the path of travel is known as **tangential acceleration**. Any acceleration vector can be split into centripetal and tangential components.

Note that for a vertical, circular loop, an object requires a minimum speed to maintain circular motion as the gravitational force is the only force that causes centripetal acceleration.

$$v = \sqrt{gr}$$

3 Work and Power

3.1 Translational Kinetic Energy

$$\text{KE} = \frac{1}{2}mv^2$$

Note different observers may observe different translational kinetic energies depending on if the observer is moving as well.

3.2 Work

Work is the amount of energy transferred in or out of a system by a force exerted by that system over a distance. Potential energies are always associated with only **conservative** (path-independent) forces. The most common *non-conservative* forces are air resistance and friction.

The non-calculus work formula $W = Fd \cos \theta$ can be extended to any path:

$$W = \int_a^b \vec{F}(r) \cdot d\vec{r}$$

This formula suggests that only the component of the force parallel to displacement can change a system's energy, and suggests that work is equal to the area under the curve of a graph of $F_{\parallel}(r)$.

The **work-energy theorem** states the change in an object's kinetic energy is equal to the sum of the work being done by all forces.

3.3 Potential Energy

The relationship between conservative forces exerted on a system and the system's potential energy is

$$\Delta U = - \int_a^b \vec{F}_{cf}(r) \cdot d\vec{r}$$

($\vec{F}_{cf}$ = conservative forces) Or, alternatively (not on the equation sheet)

$$\Delta U = -W_{cf} \text{ (potential energy is converted into work)}$$

This definition can be used to prove the potential energy equations for springs and universal gravitation:

Proof of spring energy.

$$\begin{aligned} \Delta U_{spr} &= - \int \vec{F}_{spr} \cdot d\vec{r} \\ &= - \int -kx \, dx \\ &\Rightarrow \boxed{\Delta U_{spr} = \frac{1}{2}kx^2} \end{aligned}$$

□

Proof of universal gravitational energy.

$$\begin{aligned}\Delta U_g &= - \int \vec{F}_g \cdot d\vec{r} \\ &= - \int G \frac{m_1 m_2}{r^2} dr \\ &\Rightarrow \boxed{\Delta U_g = G \frac{m_1 m_2}{r}}\end{aligned}$$

□

The integral definition of potential energy suggests that

$$F_{cf} = - \frac{dU}{dx}$$

This further suggests **stable equilibria** occur at local minima of the potential energy function (F_{cf} opposes change) while **unstable equilibria** occur at the maxima of the potential energy function (F_{cf} supports change)

3.4 Conservation of Energy

$$ME = KE + PE$$

$$\Delta ME = W_{nc} = \text{work transferred out of the system}$$

3.5 Power

$$P_{avg} = \frac{\Delta E_{transferred}}{\Delta t} = \frac{W}{\Delta t} \Rightarrow P_{inst} = \frac{dW}{dt}$$

If an object is moving at constant velocity $\vec{v}$ due to an external force $\vec{F}$, then the power transferred to the object is

$$P = \vec{F} \cdot \vec{v}$$

This usually applies to the force that a car exerts to drive at a given velocity.

4 Linear Momentum

4.1 Linear Momentum

$$\vec{p} = m\vec{v}$$

4.2 Change in Momentum and Impulse

$$\vec{F}_{net} = m\vec{a} = m \frac{d\vec{v}}{dt} = \frac{d\vec{p}}{dt}$$

Impulse is defined as follows:

$$\vec{J} = \int_{t_i}^{t_f} \vec{F}_{net}(t) dt$$

Definition 4.1 (Impulse-Momentum Theorem)

The impulse exerted on an object is equal to the object's change in momentum.

$$\vec{J} = \Delta\vec{p}$$

Proof. This is a direct result from Newton's Second Law:

$$\int_{t_i}^{t_f} \vec{F}_{net} dt = \int_{t_i}^{t_f} m \frac{d\vec{v}}{dt} dt = \int_{t_i}^{t_f} m d\vec{v} = m\vec{v}_f - m\vec{v}_i = \Delta\vec{p}$$

□

4.3 Conservation of Linear Momentum

$$\vec{v}_{cm} = \frac{\sum \vec{p}_i}{\sum m_i}$$

$$\vec{p}_{sys} = \sum \vec{p}_i$$

From the impulse-momentum theorem, in the absence of external forces (and thus impulse) the momentum of a system does not change.

From Newton's Third Law, the impulse exerted by one object on a second object is equal and opposite to the impulse exerted by the second object on the first.

4.4 Elastic and Inelastic Collisions

Definition 4.2 (Elastic Collision)

collision in which initial KE = final KE.

Definition 4.3 (Inelastic Collision)

Collision in which initial KE is not equal to final KE (lost to W_{nc}). **Perfectly inelastic** when objects stick together and move with same velocity after collision. This results in the maximal loss of energy.

5 Torque and Rotational Dynamics

5.1 Rotational Kinematics

Counterclockwise motion is considered a **positive** change in angle.

Like linear motion, there exists **angular velocity** $\omega = \frac{d\theta}{dt}$ and **angular acceleration** $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$. These definitions lead to analogous **rotational kinematic equations** for constant rotational acceleration:

$$\omega_f = \omega_i + \alpha t$$

$$\Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2$$

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$$

Note that angular velocity and acceleration are "pseudovectors" that point in the direction of the axis of rotation. *The right-hand grip rule can be used to determine the direction of these pseudovectors, but this is not needed for the exam.*

5.2 Connecting Linear and Rotational Motion

Recall the definition of arc length:

$$\Delta s = r\Delta\theta$$

This definition of arc length can be used to show the following:

$$v_T = r\omega$$

$$a_T = r\alpha$$

Note for both of these equations, they result in the **tangential component** of velocity and acceleration respectively.

5.3 Torque

Recall torque results from the perpendicular component of force relative to the position vector from the axis of rotation to the point of the force. This is mathematically represented as a cross product:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Like angular velocity and acceleration, the direction of the torque is a pseudovector, the sign indicating whether it rotates counterclockwise (positive) or clockwise (negative). Recall that torque is the rotational analog to forces.

Example 5.1

Suppose you have a bar of length L which has mass density function $m(r) = cr^2$. If the bar is fixed at $r = 0$, find an equation for the magnitude of the torque on the bar when the bar is θ below the horizontal.

Answer 5

Recall from Example 2.1 that $r_{cm} = \frac{3}{4}L$. We can consider the gravitational force as acting on the center of mass.

$$m = \int_0^L cr^2 dr = \frac{cL^3}{3}$$

$$\begin{aligned} |\vec{\tau}| &= |\vec{r}||\vec{F}_g|\sin(90 - \theta) \\ &= \frac{3}{4}L * \frac{cL^3}{3} * g \cos \theta \\ &= \boxed{\frac{cgL^4}{4} \cos \theta} \end{aligned}$$

5.4 Rotational Inertia

The **rotational inertia** of an object represents its resistance to changes in rotation, and is analogous to mass in linear motion. The rotational inertia of an object of mass m rotating at a distance r away from the axis of rotation is as follows:

$$I = mr^2$$

Because the rotational inertia of multiple objects sum, we can express it as an integral:

$$I = \int r^2 dm$$

Example 5.2

Suppose you have a rod parallel to the x axis that originates from $x = 0$ and extends to $x = L$. The rod has linear mass density $\lambda(x) = bx$ for positive constant b . Derive an equation for the moment of inertia about the line $x = \frac{L}{3}$ in terms of the total mass of the rod and the length of the rod.

Answer 6

$$\frac{dm}{dx} = bx \implies dm = bx \, dx$$

$$\begin{aligned}
 I &= \int r^2 \, dm \\
 &= \int_0^L \left(x - \frac{L}{3}\right)^2 \, dm \\
 &= \int_0^L \left(x - \frac{L}{3}\right)^2 (bx) \, dx \\
 &= b \int_0^L x^3 - \frac{2L}{3}x^2 + \frac{L^2}{9}x \, dx \\
 &= b \left[\frac{1}{4}x^4 - \frac{2L}{9}x^3 + \frac{L^2}{18}x^2 \right]_0^L \\
 &= b \left(\frac{L^4}{4} - \frac{2L^4}{9} + \frac{L^4}{18} \right) \\
 &= \frac{bL^4}{12} \\
 &= \boxed{\frac{mL^2}{6}}
 \end{aligned}$$

Note that $(x - \frac{L}{3})$ is used as r represents the distance from each dm to the parallel axis (in this case, $x = \frac{L}{3}$).

To solve for b :

$$\begin{aligned}
 m &= \int dm \\
 &= \int_0^L bx \, dx \\
 &= \left[\frac{bx^2}{2} \right]_0^L \\
 &= \frac{bL^2}{2} \\
 \implies b &= \frac{2m}{L^2}
 \end{aligned}$$

If you know an object's moment of inertia about an axis that passes through its center of mass (I_{cm}), then you can solve for the moment of inertia about any axis I' parallel to the center of mass axis with the **parallel axis theorem**:

$$I' = I_{cm} + Md^2$$

M represents the mass of the object, while d represents the distance between the two parallel axes.

5.5 Rotational Equilibrium and Newton's First Law in Rotational Form

An object is in **rotational equilibrium** if it does not have a net torque applied to it. If is in rotational equilibrium, then it maintains its current angular velocity. This is analogous to Newton's First Law in translational motion.

Note that if an object is rotationally equilibrant, this does not imply that it is translationally equilibrant, and vice versa.

5.6 Newton's Second Law in Rotational Form

$$\tau_{net} = I_{sys}\alpha_{sys}$$

6 Energy and Momentum of Rotating Systems

6.1 Rotational Kinetic Energy

$$\text{KE}_{\text{rot}} = \frac{1}{2}I\omega^2$$

$$\text{KE}_{\text{total}} = \text{KE}_{\text{rot}} + \text{KE}_{\text{translational}} = \frac{1}{2}I_{cm}\omega^2 + \frac{1}{2}mv_{cm}^2$$

6.2 Torque and Work

Torque can transfer energy in or out of a system if exerted over an angular displacement like regular work:

$$W = \int_{\theta_i}^{\theta_f} \tau \, d\theta$$

6.3 Angular Momentum and Angular Impulse

Angular momentum is represented with the symbol L . God knows why.

$$L = I\omega$$

Angular momentum for a **point** around a given axis:

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times (m\vec{v})$$

Angular impulse does not have a symbol. For this section, we will be calling it J_{ang}

$$J_{ang} = \int \tau \, dt$$

The **rotational form of the impulse-momentum theorem** suggests

$$J_{ang} = \Delta L$$

6.4 Conservation of Angular Momentum

The **total angular momentum of a system** is defined relative to **one axis**. Newton's third law suggests the angular impulse a system exerts on another system has an equal and opposite exertion on the first system.

Like the conservation of linear momentum, *angular momentum is conserved in all interactions*. If there is no net torque exerted on the system, the total angular momentum of the system will remain constant.

6.5 Rolling

Recall $\text{KE}_{\text{total}} = \text{KE}_{\text{rot}} + \text{KE}_{\text{translational}}$.

Suppose you have a rotating wheel with angular velocity $\vec{\omega}$ about an axis perpendicular to the ground. Relative to the center of mass, each point on the wheel has a tangential velocity $\vec{v}_{rot} = \vec{\omega} \times \vec{r}$.

Relative to the ground, each point on the edge of the wheel has $\vec{v} = \vec{v}_{rot} + \vec{v}_{CM}$, where $\vec{v}_{CM}$ represents the net translational velocity of the center of mass.

If the point at which the wheel touches the ground is at **instantaneous rest**, then we call this **rolling without slipping**. At that point, $|\vec{v}_{rot}| = |\vec{v}_{CM}|$ and the two vectors point in opposite directions. Thus, we have the following relation:

$$v_{CM} = v_{rot} = R\omega$$

And by proxy:

$$\Delta x = R\Delta\theta$$

$$a_{CM} = R\alpha$$

For ideal cases, rolling without slipping implies the frictional force **does not** dissipate energy from the system.

If a system is **rolling with slipping**, the system's center of mass and the system's rotational velocity cannot be related. Furthermore, rolling with slipping implies the frictional force does dissipate energy from the system.

Rolling friction is beyond the scope of Physics C.

6.6 Motion of Orbiting Satellites

Satellites are considered as objects with negligible mass relative to the object it is orbiting around.

In **circular orbit**, the system's total kinetic energy, gravitational potential energy (and thus mechanical energy), and angular momentum are constant.

In **elliptical orbit**, the system's total mechanical energy and angular momentum is constant, BUT kinetic energy and gravitational potential energy can change.

For circular path satellites, the kinetic energy can be directly related to the potential energy with the following (NOT ON THE EQUATION SHEET):

The only force applied to the satellite is the gravitational force; thus, the gravitational force must be the centripetal force.

$$G\frac{Mm}{r^2} = m\frac{v^2}{r} \implies \frac{1}{2}mv^2 = \text{KE} = G\frac{Mm}{2r}$$

Recall the potential gravitational energy formula:

$$\text{GPE} = -G\frac{Mm}{r}$$

Hence,

$$\boxed{\text{KE} = -\frac{1}{2}\text{GPE}}$$

$$\boxed{E_{tot} = \text{KE} + \text{GPE} = \frac{1}{2}\text{GPE} = -G\frac{Mm}{2r}}$$

The negative energy here suggests the satellite **lacks the energy to escape orbit**.

However, if the satellite gains enough kinetic energy, it can escape from orbit. The

precise velocity required to escape can be derived (NOT ON THE EQUATION SHEET):

An object reaches escape velocity after all its kinetic energy has been transferred to gravitational potential energy:

$$\frac{1}{2}mv_e^2 = G\frac{Mm}{r} \implies \boxed{v_e = \sqrt{\frac{2GM}{r}}}$$

With the near-surface approximation $g = G\frac{M}{r^2}$:

$$v_e \approx \sqrt{2gr}$$

r represents the radius away from the center of the given planet.

7 Oscillations

7.1 Defining Simple Harmonic Motion (SHM)

Simple Harmonic Motion (SHM) is a type of periodic (repeating) motion resulting from a *restoring force* proportional to an object's displacement from an *equilibrium position*. One common type of SHM is that caused by a spring (recall $F_{spr} = k\Delta x$) with no external forces acting on it.

7.2 Frequency and Period of SHM

Period (in the context of SHM) is defined as the time required for an object to complete one full cycle of its motion (time/cycle). Frequency is defined as its inverse: $f = \frac{1}{T}$ (cycles/time).

There exists the notion of "angular position" of SHM. The "angle" represents the "relative completion" of the cycle, with one completion being 2π . Angular frequency is defined similarly: $\omega = \frac{2\pi}{T} = 2\pi f$.

The period of an object-ideal-spring oscillator (BOTH HORIZONTAL AND VERTICAL) is given without proof:

$$T_s = 2\pi\sqrt{\frac{m}{k}}$$

The period of a simple pendulum displaced by a small angle ($\Delta\theta < 10^\circ$) is also given without proof:

$$T_p = 2\pi\sqrt{\frac{l}{g}}$$

7.3 Representing and Analyzing SHM

For an object exhibiting SHM, the displacement of that object measured from equilibrium can be represented by the equation (cos or sin both work with conversion)

$$x = x_{max} \cos(\omega t + \phi)$$

Note that changing the amplitude of a system will not change its period or frequency. Velocity and acceleration can also be derived from this equation, and can be used to show

$$v_{max} = A\omega$$

$$a_{max} = A\omega^2$$

SHM follows the following second order DE from applying Newton's second law with $a = \frac{d^2x}{dt^2}$ and $\omega^2 = \frac{k}{m}$ (see period of object-ideal-spring oscillator):

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

Physics C does not require us know how to solve the second order DE. Also, know that **resonance** exists.

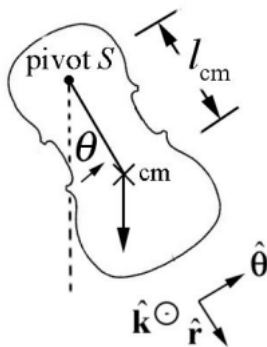
7.4 Energy of Simple Harmonic Oscillators

Conservation of energy indicates the total energy of a SHM system is constant, being constantly converted between potential and kinetic energy. The minimum energy a SHM system can express is zero (no movement). Changing the amplitude will change the energy of the system, per the spring energy equation (and thus the energy of a SHM system):

$$E_{tot} = \frac{1}{2}kA^2$$

7.5 Simple and Physical Pendulums

A **physical pendulum** is a rigid body that undergoes oscillation about a fixed axis.



The only force physical pendulums experience is gravity. Using the definition of torque and the small angle approximation $\sin \theta \approx \theta$ ($\theta < 10^\circ$) (NOT ON THE EQUATION SHEET),

$$\begin{aligned} |\tau| &= |\vec{r} \times \vec{F}_g| \\ &= |l_{cm} \hat{r} \times m\vec{g}| \\ &= -l_{cm}mg \sin \theta \\ &\approx -l_{cm}mg\theta \end{aligned}$$

Note the negative indicates that the force is restorative back towards the equilibrium angle. Applying the small-angle approximation with Newton's second law in rotational form can be used to show that this satisfies the characteristic equation of SHM in rotational form (without proof):

$$\frac{d^2\theta}{dt^2} = -\omega^2\theta$$

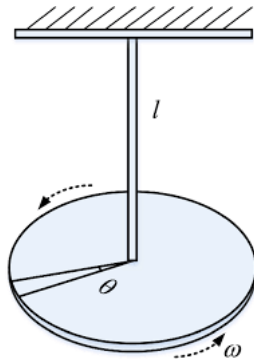
And thus, with a similar process to the derivation of angular frequency for spring SHM, this can be used to show (actually on the equation sheet)

$$T = 2\pi\sqrt{\frac{I}{mgl_{cm}}}$$

and by proxy (not on the equation sheet)

$$\omega = \sqrt{\frac{mgl_{cm}}{I}}$$

Torsional pendulums also exist. They exert a restorative force directly proportional to $\Delta\theta$: $I\alpha = -k\Delta\theta$ (and thus, $\omega = \sqrt{\frac{k}{I}}$) (neither on the equation sheet). Note k is the **torsion constant** of the wire/bar/spring/whatever.



8 Electric Charges, Fields, and Gauss's Law

8.1 Electric Charge and the Electric Force

Charge... exists. It's scalar and measured in SI units **Coulombs**.

Coulomb's Law describes the electrostatic force between two charged particles:

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} = k \frac{|q_1 q_2|}{r^2}$$

ϵ_0 represents the **electric permittivity of free space** and has a constant value (units $\text{C}^2\text{N}^{-1}\text{m}^{-2}$). Electric permittivity measures "the degree to which a material or medium is polarized in the presence of an electric field" and is determined by "the ease with which electrons can change configurations within a given material." Electric polarization can be modeled as the induced rearrangement of electrons by an external electric field.

8.2 Conservation of Electric Charge and the Process of Charging

Like charges repel. Charges on system influence each other.

8.3 Electric Fields

Electric fields exist. $\vec{E} = \frac{\vec{F}_e}{q_e} = k \frac{q}{r^2} \hat{r}$

Note while in electrostatic equilibrium (no net movement of charges), the excess charge of a conductor is distributed on the surface of said conductor, and the electric field within said conductor is 0. Note that this is **not** true for insulators: charge is distributed through the surface AND interior, and the electric field within may be non-zero.

8.4 Electric Fields of Charge Distributions

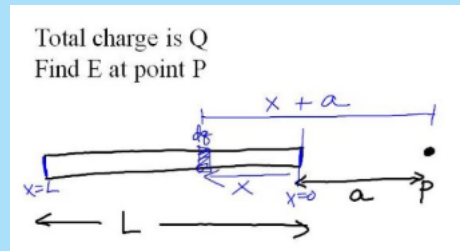
For continuous charge distributions, you can consider tiny point charges dq and apply Coulomb's Law to each of them, then add their effects together. The fact that you can add them together is known as **the principle of superposition**. Mathematically:

$$\vec{E} = k \int \frac{dq}{r^2} \hat{r}$$

You can typically directly solve for Q or express the integral in terms of dr using **linear charge density** $\lambda = \frac{dQ}{dL} \approx \frac{Q}{L}$, **surface charge density** $\sigma = \frac{Q}{A}$, or **volumetric charge density** $\rho = \frac{Q}{V}$

Example 8.1

Determine the electric field at point P.

**Answer 7**

$$\lambda dr = dQ$$

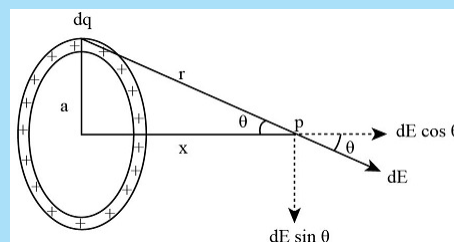
$$\begin{aligned}\vec{E} &= k \int \frac{dQ}{r^2} \hat{r} \\ &= k \hat{r} \int_a^{a+L} \frac{\lambda dr}{r^2} \\ &= \lambda k \hat{r} \left(\frac{-1}{a+L} + \frac{1}{a} \right) \\ &= \lambda k (\hat{i}) \left(\frac{L}{a(a+L)} \right) \\ &= \boxed{\frac{Qk\hat{i}}{a(a+L)}}\end{aligned}$$

Note when $a \gg L$, it degenerates to the below, which suggests that it acts as a point charge at sufficiently large distances.

$$\vec{E} \approx \frac{Qk\hat{i}}{a^2}$$

Example 8.2

Determine the electric field at P.



Answer 8

Note that "horizontal" $d\vec{E} \sin \theta$ components cancel via symmetry. Thus, you only need to consider components proportional to $\cos \theta$.

$$\begin{aligned}\vec{E} &= k \int \frac{dQ}{a^2 + x^2} \hat{i} \cos \theta \\ &= k \hat{i} \int \frac{dQ}{a^2 + x^2} \times \frac{x}{\sqrt{a^2 + x^2}} \\ &= \boxed{\frac{k \hat{i} Q x}{(a^2 + x^2)^{1.5}}}\end{aligned}$$

8.5 Electric Flux

Flux describes the amount of a given quantity that passes through a given area. **Electric flux** Φ_E describes the amount of electric field that passes through a defined area. Note the area vector is defined to be outward from a closed surface.

$$\Phi_E = \vec{E} \cdot \vec{A}$$

For an arbitrary surface:

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

8.6 Gauss's Law

Definition 8.3 (Gauss's Law (of electric flux))

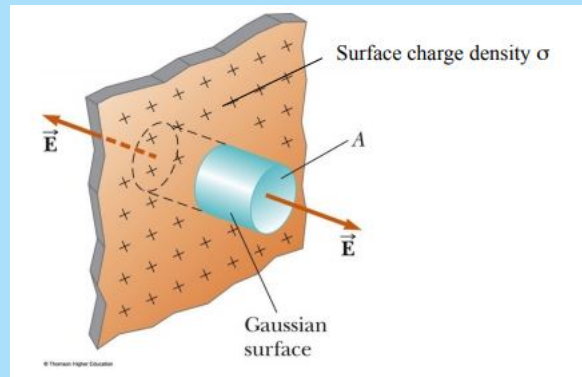
For an arbitrary **Gaussian Surface** (3d closed surface):

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

Usually you can rig the surface such that the electric field is perpendicular or parallel to all components of the area, with a sphere or a cylinder. This is also known as Maxwell's first equation.

Example 8.4

Determine the electric field surrounding an infinitely large, thin plane of positive charges with uniform surface charge density.

**Answer 9**

Set up the Gaussian surface as shown. The area vector is perpendicular to the electric field along the "wrapping" of the cylinder. The area vector is parallel to the electric field in the same direction for the two "caps" of the cylinder. Thus,

$$2EA = \frac{q_{enc}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0} \implies E = \boxed{\frac{\sigma}{2\epsilon_0}}$$

Note for the case of a two "infinite" plane capacitor, we assume the field outside the capacitor cancels. Thus,

$$E_{capacitor} = \frac{\sigma}{\epsilon_0}$$

9 Electric Potential

9.1 Electric Potential Energy

EPE of a system = amount of work required by external force to bring point charges to current positions from infinitely far away. Total EPE = sum of EPE between each pair of charged particles.

$$U_E = k \frac{q_1 q_2}{r}$$

Proof of EPE formula. Bringing q_1 to q_2 from infinity is equivalent to bringing q_2 to q_1 from infinity. EPE gained by work done to bring one charge to the given distance away.

$$\begin{aligned} \text{EPE} = W &= - \int \vec{F} \cdot d\vec{r} \\ &= - \int_{-\infty}^r k \frac{q_1 q_2}{x^2} dx \\ &= -k q_1 q_2 \left(-\frac{1}{r} - \frac{1}{\infty} \right) \\ &= \boxed{k \frac{q_1 q_2}{r}} \end{aligned}$$

□

9.2 Electric Potential

Electric potential describes the *potential energy* per unit charge at a point in space. Note it is a scalar. The electric force is **conservative**: change in EPE due to position change does not depend on the path. Electric potentials caused by different charges add.

$$V = k \frac{q}{r} \Leftrightarrow V = k \int \frac{dq}{r}$$

The electric potential difference between two points is the change in EPE between the two points:

$$\Delta V = \frac{\Delta U_e}{q} = - \int \vec{E} \cdot d\vec{r} = -Ed$$

for perpendicular distance d . The integral form comes from the work form of EPE divided by q .

Note the integral form suggests the following:

$$\vec{E}_x = -\frac{dV}{dx}$$

Also note the definition of potential suggests positive charges want to go to low electric potential (loses EPE = KE)

9.3 Conservation of Electric Energy

EPE is a type of energy, so it is conserved by the law of energy conservation.

10 Conductors and Capacitors

10.1 Electrostatics w/ Conductors

Conductors are materials in which the electrons are free to move. If they are in *electrostatic equilibrium*, then they are *stationary* within the object.

The electric field inside a conductor in electrostatic equilibrium equals zero- all the electrons (or lack thereof) exists on **solely the surface of the conductor**. If there is a charge within an empty conducting shell, then charges will gather on the internal surface to oppose that charge. By proxy, the outside of the shell will gain an opposite change in charge depending on the charge in the internal surface. **No net charge will exist within between the surfaces of the conductor.**

Furthermore, excess charges or lack thereof will equally redistribute on the surface of the conductor such that 1) the electric field outside is perpendicular to the surface and 2) the electric field AND POTENTIAL equal wherever curvature is constant (local surface charge density is at its maximum where the radius of curvature is at its minimum.) Gauss's law with a small cylinder can be used to show that the magnitude of the electric field just outside a conductor in electrostatic equilibrium is

$$E = \frac{\sigma_{local}}{\epsilon_0}$$

Anything inside a conductor in electrostatic equilibrium is shielded from external electric fields. This is known as **electrostatic shielding**.

10.2 Charge Redistribution

When two conductors touch tips, charges will redistribute such that the surfaces of the conductors have equal electric potential. Ground can be treated as a point of zero electric potential that can absorb or emit an infinite amount of charge. Charge can be induced on a conductor by grounding it in the presence of an external electric field.

10.3 Capacitors

A **parallel plate capacitor** consists of two separated parallel conducting plates that can hold equal amounts of charge with opposite signs. *Capacitance* relates the magnitude of the charge stored on the plates vs the voltage difference between them, and is dependent on the physical properties of the capacitor:

$$C = \frac{Q}{\Delta V} = \frac{\kappa \epsilon_0 A}{d}$$

The **dielectric constant** κ comes from the insulating material between the two plates. See 10.4 for more.

Because it takes work to move charges from one plate to the next as the charge difference builds, there is a potential energy associated with the capacitor:

$$W = q\Delta V \implies dW = \Delta V dq$$

$$\begin{aligned}
W &= \int_0^Q \Delta V \, dq \\
&= \int_0^Q \frac{q}{C} \, dq \\
&= \boxed{\frac{Q^2}{2C} = \frac{1}{2} Q \Delta V}
\end{aligned}$$

If the gap between the plates is small relative to their size, then you can assume the plate is "infinitely large" and apply Gauss's law. With the assumption that the electric field outside the other side is zero, the electric field can be shown to be

$$E = \frac{Q}{\epsilon_0 A}$$

10.4 Dielectrics

In a dielectric material, electrons cannot flow as easily in a material; the material becomes polarized in the presence of an external electric field. The dielectric constant is defined relative to the electric permittivity of the field (note the second definition is not on the equation sheet)

$$\kappa = \frac{\epsilon}{\epsilon_0} = \frac{E_0}{E}$$

11 Electric Circuits

11.1 Electric Current

Current is the rate at which charge passes through a given cross-sectional area of wire.

$$I = \frac{dq}{dt}$$

Current can also be modeled as the movement of **charge carriers** travelling through the conductor with a derived **drift velocity**. The equation $I = nqv_dA$ shows this (on the equation sheet), with n = charge carrier density (units m^{-3}), q = charge per charge carrier (units C), v_d representing average drift velocity in metres per second, and A representing cross-sectional area. nq is often given together to cancel out charge carriers.

The quantity nqv_d can also be modeled as **current density** $\vec{J} = nq\vec{v}_d$ (units Cm^{-3}), related to current with the following:

$$I = \int \vec{J} \cdot d\vec{A}$$

A potential difference across a conductor (the emf ϵ) creates an electric field within the conductor that is proportional to the resistivity of the conductor and the current density:

$$\vec{E} = \rho\vec{J}$$

11.2 Simple Circuits

Remember that all current on diagrams are drawn using conventional current (movement of "positive charges")

11.3 Resistance, Resistivity, and Ohm's Law

$$R = \frac{\rho l}{A} \implies R = \int \frac{\rho(l)}{A} dl$$

11.4 Electric Power

$$W = q\Delta V \implies P = \frac{dq}{dt}\Delta V = I\Delta V = I^2R = \frac{V^2}{R}$$

Note last two are more used for power dissipated.

11.5 Compound DC circuits

Recall that voltmeters are attached parallel to the circuit. Also recall that non-ideal batteries can be modelled as an ideal battery in series with an "internal resistance" resistor for net terminal voltage (not on eq sheet)

$$\Delta V_{\text{terminal}} = \epsilon - Ir$$

11.6 Kirchoff's Loop Rule

Kirchoff's Loop Rule is a consequence of the conservation of energy.

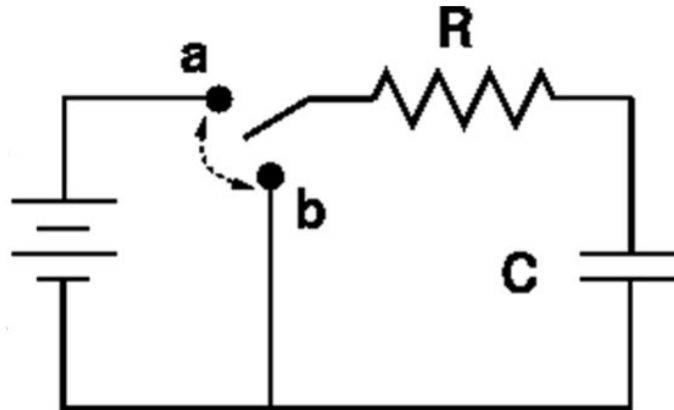
11.7 Kirchoff's Junction Rule

Kirchoff's Junction Rule is a consequence of the conservation of charge.

11.8 Resistor-Capacitor Circuits

(As a result of the conservation of charge, each of the capacitors in series must have an equal magnitude of charge on each plate.)

A **Resistor-Capacitor (RC)** circuit is a circuit with a resistor and capacitor in series.



Using Kirchoff's loop rule with the battery part of the circuit leads to the following:

$$0 = \epsilon - IR - \Delta V_{\text{capacitor}} \implies \epsilon = \frac{dq}{dt}R - \frac{q}{C}$$

This is a first-order differential equation that can be solved for charge in terms of time:

$$q(t) = Q(1 - e^{-\frac{t}{RC}})$$

The **time constant** of the circuit $\tau = R_{eq}C_{eq}$ represents the time for a charging capacitor (battery connected) to increase to 63% of its final value and the time for a discharging capacitor (battery disconnected) to decrease by 37% of its initial value.

For an uncharged capacitor being connected to a battery, initially it acts as a wire as it charges, but eventually builds up resistance until it stops conducting charge. A discharging capacitor acts as a battery initially, but as it discharges down it decreases in emf.

12 Magnetic Fields and Electromagnetism

Magnetic field lines must form closed loops, per **Gauss's law of Magnetism** (Maxwell's Second Equation):

$$\oint_C \vec{B} \cdot d\vec{A} = 0$$

Magnetic dipoles result from the circular or rotational movement of electrons, either up or down. Recall that **poles always occur in pairs**.

Magnetic permeability μ is a measurement of the amount of magnetization in a material in response to an external magnetic field. Free space a constant value μ_0 .

12.1 Magnetism and Moving Charges

Recall the movement of charges leads to magnetic fields. The magnitude of the magnetic field is some sort of a cross product between the velocity of the object and the position vector.

Similarly, magnetic fields induce a magnetic force on charged objects moving within that field given as follows:

$$\vec{F}_b = q(\vec{v} \times \vec{B})$$

This usually leads to circular motion, as the magnetic force is always perpendicular to the velocity vector.

"The Hall effect describes the potential difference created in a conductor by an external magnetic field that has a component perpendicular to the direction of charges moving in the conductor."

12.2 Magnetic Fields of Current-Carrying Wires and the Biot-Savart Law

The **Biot-Savart law** describes the magnitude and direction of a magnetic field created by an electric current:

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I(d\vec{l} \times \hat{r})}{r^2}$$

Example 12.1

Use the Biot-Savart Law to derive the magnetic field caused by a circular wire of radius r carrying current I at the center of the circle..

Answer 10

Intergrate the BS Law and note that $\frac{\mu_0}{4\pi}$, I , and r^2 are constant:

$$\vec{B} = \frac{\mu_0 I}{4\pi r^2} \int d\vec{l} \times \hat{r}$$

Note that the unit vector pointing from each differential length is perpendicular to the tangent position vector. Therefore, the intergral collapses to

$$|\vec{B}| = \frac{\mu_0 I}{4\pi r^2} * 2\pi r = \boxed{\frac{\mu_0 I}{2r}}$$

The right-hand curl rule can be used to determine the direction of the vector (parallel to the area vector)

Also note a magnetic field will exert a force on a current carrying wire, a collection of moving charges:

$$\vec{F}_B = \int I(d\vec{l} \times \vec{B})$$

12.3 Ampere's Law

Ampere's Law relates the magnitude of the magnetic field to the current enclosed by an **Amperian loop**.

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$$

Definition 12.2 (Amperian loop)

A closed line loop in 3d space. "Enclosed current" refers to the current of any wire that passes through with a parallel component to the area vector of the Amperian loop.

Example 12.3

Use Ampere's Law to determine the magnitude of the magnetic field caused by a straight, current-carrying wire a distance of r away.

Answer 11

The magnetic field generated by a current-carrying wire is tangent to concentric circles of radius r . Using an Amperian loop of a circle with radius r , the dot product is always parallel and therefore

$$B(2\pi r) = \mu_0 I_{enc} \implies B = \frac{\mu_0 I_{enc}}{2\pi r}$$

The direction of the magnetic field vector can be found using the right-hand grip rule.

Example 12.4

Use Ampere's Law to determine the magnitude of the magnetic field caused by a long solenoid of length L and N loops.

Answer 12

The magnetic field generated by a long solenoid is parallel to the "area vectors" of the solenoid loops. Suppose an Amperian loop of a rectangle, with one long side inside the solenoid and one outside. Solenoids have approximately no magnetic field outside the solenoid. The shorter edges of the loop are parallel to the magnetic field, so by the cross product they evaluate to zero. Therefore, the integral degenerates to

$$B(L) = \mu_0 N I$$

Note the N results from N loop wires piercing the Amperian loop. Recall that n is the loop count per unit length. Therefore,

$$B = \mu_0 n I$$

You need to know that Ampere's law is not fully accurate, as it does not account for *magnetic fields caused by changing electric fields* (because yeah they do that). This is codified in **Maxwell's fourth equation** (which you do NOT need to know) (need line integral)

$$\int \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

13 Electromagnetic Induction